Golla Karthik Yadav

ee23btech11024@iith.ac.in | gollakarthikyadav7@gmail.com | +91 8328544792 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.47
SRI CHAITANYA JNR KALASALA , MADHAPUR,HYDERABAD,TELANGANA	Class XII (Telangana State Board of Intermediate Education)	2023	98.00%
SRI CHAITANYA HIGH SCHOOL,KUKATPALLY, MEDCHAL DISTRICT	Class X (Telangana State Board of Secondary Education)	2021	10.00%

WORK EXPERIENCE

Deccan AI -- AI Data Training & Evaluation Freelancer:

- Worked across **4** AI training projects since Mar 2026, evaluating LLM outputs across coding, reasoning, SQL, and human interpretability tasks to provide high quality feedback for model post training.
- Performed structured evaluation of LLMs and AI agents across reasoning, coding, tool use, and task completion, identifying errors and ranking responses based on task specific quality criteria.
- Curated and refined human generated training data through annotation, response improvement, and multi stage quality checks, supporting iterative model training and evaluation workflows.

SKILLS

- **Programming Languages :** C , C++ , Python(NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch) , SQL , MATLAB.
- **Tools :** LT Spice , nanoHUB , Solid Edge , Fusion360.
- **Others :** Linux , Bash/Shell ,Data Structures and Algorithms , LaTeX , Git , Github , Jupyter Notebook , Verilog
- **Soft Skills :** Team Management , Communication , Decision Making , Adaptability , Time Management.

PROJECTS

Low-Latency Limit Order Book & Matching Engine | C++20, Linux :

- Engineered a **deterministic price-time-priority matching engine** supporting limit/market orders, partial fills, cancellation, replacement, and **multi-instrument order books**.
- Implemented a **pre-allocated order pool**, **custom open-addressing order index**, and **intrusive FIFO price-level queues**, enabling **O(1) order cancellation** while minimizing hot-path dynamic allocation.
- Built a **lock-free SPSC pipeline** using C++ atomics with **acquire/release memory ordering** and developed **p50/p95/p99/p99.9 tail-latency benchmarks** with sanitizer-backed testing.

Market Data Feed Handler & Replay Engine | C++20, Linux :

- Built a **Linux market-data pipeline** using **non-blocking UDP sockets and epoll**, with explicit binary serialization/deserialization and **network-byte-order parsing** for a custom exchange-style feed protocol.
- Implemented **sequence validation** and feed-health handling for **gaps, duplicates, out-of-order messages, and malformed packets**, while maintaining **local multi-instrument order books**.
- Developed **lock-free SPSC processing**, **binary feed capture**, deterministic replay, and **golden-file regression testing** for reproducible market-data processing and end-to-end validation.

Communication Efficient Distributed Learning for Convex Logistic Regression :

- Developed a communication efficient distributed logistic regression framework across **100 wireless edge nodes,** optimizing a global convex objective through decentralized local training and centralized weight aggregation.
- Formulated UE participation as a convex optimization problem under wireless fading and transmission constraints, deriving a **KKT-based trial-allocation strategy** to maximize the joint probability of successful model updates.
- Implemented numerical bisection with integer-aware trial allocation across 100 nodes, evaluating logistic regression

on MNIST and analyzing model accuracy against communication rounds under a **50m wireless network setup**.

RAG Document Q&A Chatbot :

- Engineered a **multi format RAG pipeline** for document Q&A, combining Sentence Transformers embeddings, **FAISS vector search**, and **Gemini models** to deliver **context grounded responses** from user uploaded documents.
- Implemented **300 word overlapping chunking** and **top 4 semantic retrieval** over indexed document embeddings, enabling targeted context selection for downstream LLM generation.
- Developed an interactive Gradio interface supporting **multi file PDF/TXT/CSV ingestion**, automated indexing, conversational querying, and source aware responses.

POSITIONS OF RESPONSIBILITY

Events Coordinator | Milan 2025

- Coordinated planning, logistics, participant management, and on-ground execution for the **college-wide fest**.
- Collaborated with organizing teams to ensure smooth and timely execution of multiple events.

Events Coordinator | Diesta 2026

- Coordinated events for an **inter-departmental** fest involving participation from all college departments.
- Managed event logistics, scheduling, participant coordination, and cross-departmental collaboration.

Mess Monitoring Council | Mess Inspector

- Inspected daily mess operations, ensuring food quality, hygiene, cleanliness, and service standards.
- Inspected mess facilities, identified operational issues, and coordinated with council members and mess staff to address concerns.

RELEVANT COURSES

Programming Courses : Introduction to Programming , Introduction to AI , Introduction to Modern AI , Pattern Recognition and Machine Learning , Data Structures and Applications , Convex Optimization.

Math Courses : Calculus , Matrix Theory , Probability and Random Processes , Complex Variables , Vector Calculus , Differential Equations , Predictive Modelling.

Core Courses : Digital Systems , Signals and Systems , Digital Signal Processing , Pattern Recognition and Machine Learning , Control Systems , Analog Electronics , Semiconductor Device Fundamentals , Power System Analysis , Electrical Machines , Statistical Inference , Information Theory.

EXTRACURRICULAR

- **Competitive & Recreational Activities:** Active interest in **Chess, Badminton, Table Tennis, PC Gaming, and Mobile Gaming**, with a strong focus on strategic and competitive gameplay.
- **Silver Medalist in Call of Duty Mobile (Team Event)** at the **Milan Inter-Hostel Competitions**, representing the hostel team.